Submission Worksheet

Submission Data

Course: IT202-008-S2026
Assignment: IT202 Module 4 SQL Challenge
Student: Kurt C. (kc784)
Status: Submitted | **Worksheet Progress:** 100+%
Potential Grade: 11.00/10.00 (110.00%)
Received Grade: 0.00/10.00 (0.00%)
Started: 3/26/2026 5:32:54 PM
Updated: 3/26/2026 5:32:54 PM
Grading Link: https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-module-4-sql-challenge/grading/kc784
View Link: https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-module-4-sql-challenge/view/kc784

Instructions

- Overview Link: https://youtu.be/oW7AK39mgmo

1. Ensure you read all instructions and objectives before starting.
2. Create a new branch from qa called `M4-Homework`
    1. `git checkout qa` (ensure proper starting branch)
    2. `git pull origin qa` (ensure history is up to date)
    3. `git checkout -b M4-Homework` (create and switch to branch)
3. Copy the template code from here: GitHub Repository - M4 Homework
    - Put all into an `M4` folder inside your `public_html`
    - Expected files
        - [ ] sql/000_create_table_todos.sql
        - [ ] sql/init_db.php
        - [ ] todos/create.php
        - [ ] todos/pending.php
        - [ ] todos/completed.php
        - [ ] nav.php
        - [ ] two index files (M4 and todos)
    - Immediately record to history
        - [ ] `git add public_html`
        - [ ] `git commit -m "adding M4 HW baseline files"`
        - [ ] `git push origin M4-Homework`
        - [ ] Create a Pull Request from `M4-Homework` to qa and keep it open
4. Fill out the below worksheet
    - Each Problem requires the following as you work
        - [ ] Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
        - [ ] Update ucid in nav.php where noted
        - [ ] Code solution (add/commit periodically as needed) (refer to comments in the create, pending,

completed files)

5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
    1. `git add .`
    2. `git commit -m "adding PDF"`
    3. `git push origin M4-Homework`
    4. On Github merge the pull request from `M4-Homework` to `qa`
    5. On Github create a pull request from `qa` to `prod` and immediately merge. (This will trigger the `prod` deploy to make the render.com prod links work)
7. Upload the same PDF to Canvas
8. Sync Local
    1. `git checkout qa`
    2. `git pull origin qa`

Section #1: (1 pt.) Prep

Progress: 100%

Details:
Once the files are copied in, run your local PHP server and visit `/M4/sql/init_db.php` in the browser to trigger the table generation.

Task #1 (0.50 pts.) - Show the output of the init_db.php tool

Progress: 100%

Details:
The output should note either successfully executed or skipped (if you ran this already)

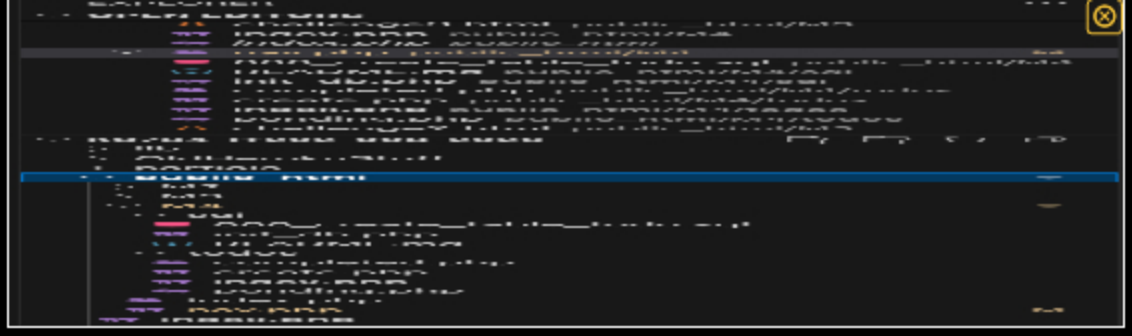


successful

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Task #2 (0.50 pts.) - Show the project view of your editor to show all files are present

Progress: 100%



project view of files

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Section #2: (3 pts.) Create Page

Progress: 100%

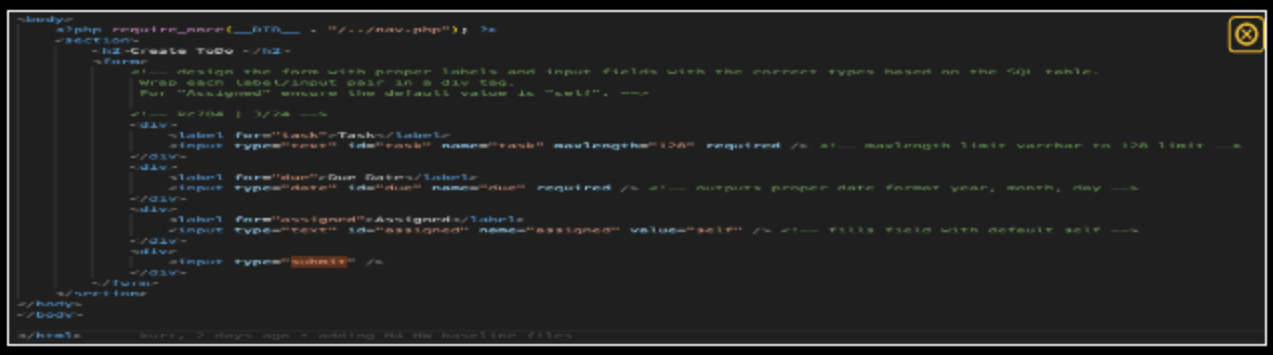
Task #1 (0.75 pts.) - HTML Form Challenge

Progress: 100%

Part 1:

Progress: 100%

Details:
Show the code snippet related to the completed HTML form



html form

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Part 2:

Progress: 100%

Details:
Items

- Note the field names and why the specific types were chosen
- Also note how you made "self" the default for the assigned input

Your Response:

With task using type="text", it stores a varchar string, due using type="date" to match the DATE column with proper formatting, and assigned using type="text" as it also deals with varchar string. The naming matches what each $_GET["task"/"due"/"assigned"] need. In terms of the "self" default with value="self", it set on the given inputs so that it is pre-filled unless the user makes a change.

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Task #2 (0.75 pts.) - Form Submission PHP Validations

Progress: 100%

Part 1:

Progress: 100%

Details:

- Validate the incoming data for correct format based on the SQL table definition.
- When not valid, provide a user-friendly message of what specifically was wrong and set `$is_valid` to false
- Assigned should check for "self" if a valid format/value isn't provided; it should default to "self" if all checks for this property fail
- Include screenshot(s) from Render.com QA demonstrating each validation (ensure URL is visible)



empty task



task name too long





invalid date



assigned is default "self"

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Part 2:

Progress: 100%

Details:
Briefly explain your validation logic.

Your Response:

All data within the form is sent through GET and PHP validates each field before adding the data. Each task is trimmed from extra whitespaces and checks if they're empty, and if empty, $is_valid is flagged to false and sends an error. On the other hand, if the length of the task is longer than 128 chars, an error is also sent. For the due date, an error is sent if that field is left empty and has to match the year,month, date format in MySQL. And assigned is default to "self" unless the user changes it.

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Task #3 (0.75 pts.) - Show the code snippet related to the INSERT query

Progress: 100%

Part 1:

Progress: 100%

Details:

Design a query to insert the incoming data to the proper columns. Ensure valid and proper PDO named placeholders are used. https://phpdelusions.net/pdo



inserted new todo



seen in pending todos

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Part 2:

Progress: 100%

Details:
Briefly describe `how` the query works in plain words.

Your Response:

The query works by inserting a new row into M4_Todos table and under the names of task, due, and assigned instead of putting them into the query string immediately. The PDO then switches each of the task, due, and assigned placeholders with the respective values and it helps keep user input separate from the SQL.

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Task #4 (+ 1.01 pts.) - Extra Credit: Handle Unique Constraint Errors

Progress: 100%

Part 1:

Progress: 100%

Details:
Show the code related to handling the unique constraint errors and the user-friendly message

```
        } catch (PDOException $e) {
            // kc784 | 3/24
            // extra credit
            // check if the exception was related to a unique constraint
            // provide an appropriate user-friendly message for this scenario
            // Otherwise show the default message below
            if ($e->getCode() == 23000) { // unique constraint
                echo "A todo with that task + due date already exists."; // notifies user of already existing task + date
            } else {
                echo "There was an error inserting the record; check the logs (terminal)"; // generic message
            }
            error_log("Insert Error: " . var_export($e, true)); // shows in the terminal
        } <-- #67-79 catch (PDOException $e)
    } else {
        error_log("Creation input wasn't valid");
    }
} <-- #9-83 if (empty($diff))
```

related code for constraint errors

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Part 2:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

The code checks the error code using the $e->getCode() and if it is 23000, the unique constraint was violated. In that instance, a task with the same exact date combination is already in the table and that message is shown to the user. Other errors fall under the else and gives a generic error message to the user. The full details are sent to the terminal so that typical users aren't given access to sensitive information.

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Task #5 (0.75 pts.) - Create Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if not prod

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/blob/M4-Homework/public_html/M4/todos/create.php

URL
https://github.com/kurtjcastr

URL #2
https://kc784-it202-008-prod.onrender.com/M4/todos/create.php

URL
https://kc784-it202-008-prod.

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Section #3: (3 pts.) Pending Page

Progress: 100%

Task #1 (1 pt.) - Fetching ToDos

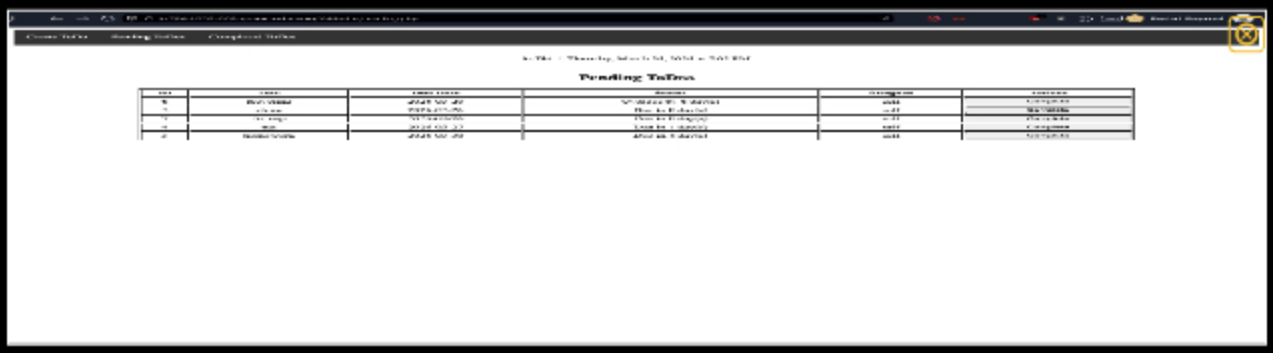
Progress: 100%

Part 1:

Progress: 100%

Details:

- Refer to the HTML table in the given code and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table, aside from the notes below
- For the Status part, you'll need to calculate the "days_offset" from the due date, ensure the virtual column matches "days_offset"
- For Actions, this isn't part of the query, and there's nothing special to select for it
- Filter the results where the todo item is NOT completed and order the results by those due soonest
- No limit is required
- Include screenshot(s) from Render.com QA demonstrating 5 different records (ensure URL is visible)



5 records

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Part 2:

Progress: 100%

Details:
Briefly describe `how` the query works in plain words.

Your Response:

The query goes over the columns of the table headers of id, task, and due in M4_Todos and the DATEDIFF calculates the difference between the due date and current date which labels the difference as days_offset. The assigned is selected last and WHERE is_complete=0 filters out anything that has been completed by ascending order through ORDER BY due ASC.

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Task #2 (1 pt.) - Completing ToDos

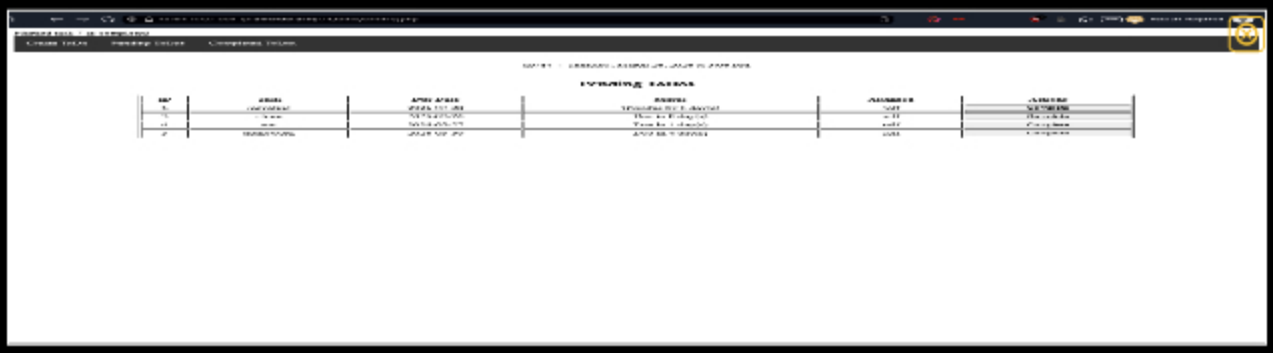
Progress: 100%

Part 1:

Progress: 100%

Details:

- Create a query that'll update the respective ToDo, marking it complete and setting the date for the completed date field as today
- Ensure the "id" is utilized using proper PDO named placeholders so that only the one item is updated
- Add an extra clause to update only if the complete field of the record is not set



completed fat nap id task 7 message in pending



completed fat nap id task 7 message in completed

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Part 2:

Progress: 100%

Details:
Briefly describe `how` the query works in plain words.

Your Response:

The query updates a single row into the M4_Todos by setting the value to 1 of is_complete and completed to NOW(). WHERE id also makes sure that only that todo task that was selected is updated and the extra AND is_completed=0 is used as a safety net in case a task has already been marked complete so no further updates are made by the query.

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Task #3 (1 pt.) - Pending Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from GitHub (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if not prod

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/blob/M4-Homework/public_html/M4/todos/pending.php

URL
https://github.com/kurtjcastr

URL #2
https://kc784-it202-008-prod.onrender.com/M4/todos/pending.php

URL
https://kc784-it202-008-prod.

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Section #4: (2 pts.) Completed Page

Progress: 100%

Task #1 (1 pt.) - Fetching ToDos

Progress: 100%

Part 1:

Progress: 100%

Details:

- Refer to the HTML table below and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table, aside from the notes below
- For the completed date, you'll need to extract the date portion from the completed column
- For the Status part, you'll need to calculate the "days_offset" from the completed date, ensure the virtual column matches "days_offset"
- Filter the results where the todo item is completed and order the results by most recently completed and most recently due
- No limit is required
- Include screenshot(s) from Render.com QA demonstrating 5 different records (ensure URL is visible)



5+ records completed

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Part 2:

Progress: 100%

Details:
Briefly describe `how` the query works in plain words.

Your Response:

The query selected the columns within the M4_Todos and DATE(completed) trims the time, returning only the date. DATEDIFF then calculates the difference in days from the due date and the completed date and names the result days_offset then assigned. The completed tasks are then filtered by WHERE is_complete=1 and is sorted by the recently completed at the top through ORDER BY completed DESC, due DESC.

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Task #2 (1 pt.) - Completed Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from GitHub (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if not prod

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/blob/M4-Homework/public_html/M4/todos/completed.php

URL
https://github.com/kurtjcastr

URL #2
https://kc784-it202-008-prod.onrender.com/M4/todos/completed.php

URL
https://kc784-it202-008-prod.

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Section #5: (1 pt.) Misc

Progress: 100%

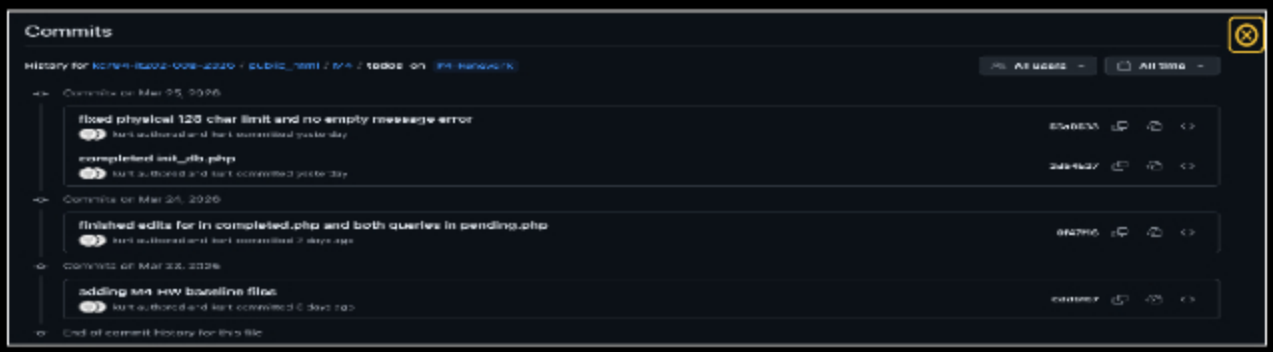
Task #1 (0.33 pts.) - Github Details

Progress: 100%

Part 1:

Progress: 100%

Details:
From the Commits tab of the Pull Request screenshot the commit history



commit history / completed init_db.php should've been create.php

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Part 2:

Progress: 100%

Details:
Include the link to the Pull Request (should end in `/pull/#`)

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/pull/3

URL
https://github.com/kurtjcastro/kc

URL #2
https://github.com/kurtjcastro/kc784-it202-008-2026/pull/7

URL
https://github.com/kurtjcastro/kc

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Task #2 (0.33 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click `Projects` and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary



overall time



file time

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Task #3 (0.33 pts.) - Reflection

Progress: 100%

Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

I learned how to use PDO and to properly write queries with safe parameters, and how to handle specific database errors, such as unique constraints.

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Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

The easiest part of the assignment was creating the HTML form because it was a simple and straightforward process of formatting SQL columns with specific input types.

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Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Briefly answer this question (at least a few decent sentences)

Your Response:

The challenging part of the assignment was definitely getting the offset correct because there were different points of reference that I needed to calculate correctly in order to get an accurate difference between the completed and current date.

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